

Assignment 3

CSED441 - Introduction to Computer Vision

Due: December 20th (Sat), 23:55 KST

Final Term Project Objective:

Build an image classification model and maximize its performance.

The Final Term Project will be conducted as a **Kaggle Competition** (Google Colab is optional).

[Competition link](#)

Competition Policy

- Only CNN models are permitted.
- Predictions must be generated solely by the model; manual labeling by a human is not allowed.
- Strict compliance with the academic honor code is required; any form of plagiarism is strictly prohibited.

Submission Policy

- All students must change their nickname to their student ID (go to the "Team" tab above)
- After submitting your solution on Kaggle, log into PLMS (at POSTECH) and submit the following:
 1. Your final ipynb file.
 2. The final report, which describes the techniques implemented to improve the model's classification performance. Name your report as a single PDF file named: assn3_[your student ID].pdf (e.g., assn3_report_20251234.pdf)
 3. After completing the assignment, re-compress the directory back into a zip file named: assign3_studentID.zip (e.g., assign3_20251234.zip)

Rubric

Your final submission will be evaluated based on the criteria outlined below. The report must clearly detail your design choices and provide concrete evidence of performance improvement from the baseline model.

I. Minimum Performance Threshold (25 points)

You will earn 25 points if your final model meets the minimum required performance threshold.

- Minimum performance threshold: **61**
- If your model fails to meet this threshold, the score for this component will be 0 points.

II. Ranking in Leaderboard (25 points)

Up to 25 points will be awarded based on your final public leaderboard rank. Assuming approximately 40 participating teams, the score is mapped from rank r (where $r=1$ is the best) as follows:

Rank (r)	Points
1	25
2	23
3	21
4	19
5	18
6–8	16
9–12	14
13–20	10
21–30	5
31–40	0

III. Final Report (50 points)

You are required to conduct ablation studies on at least two of the following components. Each component's maximum contribution is indicated in parentheses:

- Architecture Search (10): e.g., model, BatchNorm, stacking layers, etc.
- Data Augmentation (20): e.g., RandomFlip, RandomRotation, ColorJitter, CutMix, MixUp, etc.
- Optimizer / Training Strategy (20): e.g., optimizer, learning rate schedulers, decay, warmup, etc.

Ablation results must be clearly summarized with supporting evidence (e.g., plots) for full credit.

The report should explicitly state:

- The design decision/change made.
- The effect on performance compared to the baseline model.
- A brief interpretation of the results.

If you have any questions, please contact TAs:

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